Thresholds Survive, Products Collapse: Model Family Governs Robustness to Perceptual Coding in Speech Emotion Recognition

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Abstract

Speech emotion recognition (SER) is validated on uncompressed audio and deployed on coded audio. We measure what that substitution costs and find the answer is not a property of the codec. Fifteen classifiers — fourteen scikit-learn models and a neural network — are trained on 436 named acoustic descriptors from CREMA-D under a speaker-independent split, then served MP3 and MP4/AAC at 64 kbps. Under MP3 the outcome is categorical rather than graded: every model that thresholds a feature value stays within 1.4 points of its clean balanced accuracy, while every model that multiplies one loses 11.7 to 38.3 points. The two families do not overlap. Because the features are named the mechanism is legible: perceptual bit allocation abandons the top octave, and one descriptor of that octave arrives 38.6 training standard deviations out of range. Masking that single column recovers 97.9 % of an RBF-SVM's loss and 92.3 % of the network's at no cost on clean audio, so the failure is covariate shift on an identifiable channel rather than lost emotional information. Attribution stability follows the same family split, and we retain two negative results constraining how such stability should be used.

Index Terms: speech emotion recognition, perceptual audio coding, covariate shift, explainable AI, model robustness

1. Introduction

Audio rarely reaches an inference endpoint in the representation in which it was captured. Contact-centre archives hold MP3, streaming clients negotiate AAC or Opus, and a clip may be transcoded several times between microphone and model. Evaluation practice runs the other way: SER corpora are distributed as uncompressed WAV and read directly from disk.

The gap is usually dismissed on a reasonable-sounding argument. Any ingestion path decodes to linear PCM before feature extraction, so the delivery format is an implementation detail. Where the assumption has been tested, it has been tested as an average: Reddy and Vijayarajan [1] report codec effects on SER that are real but modest and front-end-dependent.

We show that the average is the wrong summary, because the underlying distribution is bimodal. Training fifteen models on one feature table, one split, and one set of labels, and serving them MP3 at 64 kbps, produces a clean partition: seven models lose essentially nothing and eight lose between a fifth and two thirds of their headroom above chance. Nothing about the codec distinguishes the two groups — they receive bit-identical inputs. What distinguishes them is how the decision rule consumes a feature value. A tree asks whether a descriptor exceeds a threshold, and the answer is unchanged whether it reads 16.4 or 46.4; a linear, kernel, or neural model multiplies that excursion straight into its decision function.

Because the 436 features carry acoustic names, the argument does not stop at the observation. We locate the contaminated channel, verify it is the one the encoder abandons, repair the failure by masking it, and confirm on clean audio that the masked column was never load-bearing evidence.

Contributions. We report a model-family split in codec robustness under MP3 with no overlap between the two families and a mechanism stated in acoustic terms (§4.3); a diagnosis-and-repair procedure that recovers almost the entire collapse from a single column, with a clean-audio control ruling out the trivial explanation (§4.5); cross-format attribution stability for six model families on identical inputs, including two negative results retained rather than smoothed (§4.4); and a container control confirming that the operational risk localises entirely to the codec.

2. Related Work

Coding and speech tasks. Degradation of speech systems under lossy coding is documented but shallowly characterised. The most consequential result comes from neural codec benchmarking: Wu et al. [2] show that signal-domain fidelity metrics fail to predict retention of task-relevant information, so a codec can score well under reconstruction error while discarding paralinguistic content. We therefore report the induced feature shift and the model behaviour as separate quantities.

Interpretability. Post-hoc attribution rests on a small set of workhorses: local surrogates [3], Shapley-value attribution [4] with an exact algorithm for tree ensembles [5], and for differentiable models path-integral [6] and reference-based [7] decompositions, available in consolidated form [8]; surveys map the space [9]. Rudin [10] argues that for high-stakes decisions the post-hoc layer should be replaced by an intrinsically interpretable model, a position our §4.2 result speaks to directly and unfavourably. Transfer of these methods to audio is contested: Haunschmid et al. [11] argue that transplanting image-segmentation logic onto time–frequency representations yields components with no perceptual correlate, and related critiques extend to music and to speech emotion specifically [12, 13, 14]. We sidestep the objection by attributing over named acoustic descriptors in the tradition of GeMAPS [15] and openSMILE [16], where every unit of attribution has a phonetic reading by construction.

Attribution fragility. Attributions can be substantially altered by perturbations below the threshold of perception while predictions remain stable [17, 18]; some saliency methods are statistically independent of model parameters [19], and different explanation methods routinely disagree on the same model [20]. We test both failure modes directly. That prior work uses adversarially constructed perturbations; a transcode is not adversarial. It is routine, ubiquitous, and nobody logs it.

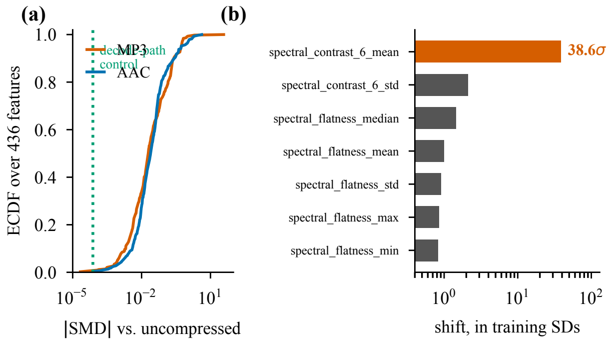

Figure 1: What the codec does to the feature table, before any model is involved. (a) Empirical CDF of |SMD| over all 436 features against uncompressed audio, 7,441 clips: both codecs leave the median feature untouched and differ only in the tail. The green line marks the decode-path control median (7.6×10^{-5}). (b) The MP3 tail in *training* standard deviations: one descriptor of the 6–8 kHz band moves 38.6σ, the next largest 2.1σ.

3. Method

3.1. Corpus and coding conditions

We use CREMA-D [21]: 7,442 clips from 91 actors, each delivering one of twelve fixed sentences in one of six acted emotions (anger, disgust, fear, happiness, neutral, sadness); 5.26 h of audio. The fixed lexical inventory is load-bearing — holding the words constant across emotions removes the lexical channel as a source of discriminative variance, so any recovered signal must be carried by acoustic realisation.

Every clip is rendered into four conditions descended from a single canonical reference x (16 kHz mono 16-bit PCM, EBU R128 normalised to $I = -23$ LUFS, TP $= -2$ dB): `ref` (x itself); `mp3_64` (libmp3lame, 64 kbps); `mp4_aac64` (AAC, 64 kbps, audio-only MP4); and `roundtrip_wav`, the AAC output decoded back to PCM through one identical FFmpeg call. Single ancestry makes the format contrast the only systematic difference between conditions, and a fixed decode path keeps the decoder from confounding the codec.

`roundtrip_wav` is the container control: codec output held fixed while the delivery wrapper changes, so any difference from `mp4_aac64` is attributable to routing rather than coding. Its median standardised feature difference from `mp4_aac64` is 7.6×10^{-5}, with only 12 of 436 features exceeding 0.05 — the noise floor of the study. Any effect larger is a codec effect.

3.2. Features

Each file is reduced to 436 named descriptors, every frame-level contour collapsed to eight order statistics (mean, median, SD, IQR, min, max, skew, kurtosis). The families are *spectral* (114: centroid, bandwidth, rolloff, flatness, flux, entropy, slope, ZCR, RMS, seven-band spectral contrast, eight band-energy ratios, and explicit high-frequency survival ratios above 4 and 6 kHz), *cepstral* (240: 20 MFCCs with Δ and $\Delta\Delta$), *prosodic* (56, computed in Praat via Parselmouth [22]: F_0 statistics and slope, jitter, shimmer, HNR, formants F_1–F_5 with bandwidths, intensity, voiced-segment timing and cepstral peak prominence), *chroma* (24) and two global descriptors. Spectral and cepstral features use librosa [23] at 512-point FFT, 160-sample hop. Chroma tuning is pinned to zero rather than estimated, because an estimated tuning could itself shift under compression and would silently become part of the effect being measured.

Extraction over $7{,}442 \times 4$ files failed on exactly one clip in all four conditions, `1076_MTI_SAD_XX`, which the corpus documentation lists as having no audio; 7,441 clips form the analysis set.

3.3. Models and protocol

Actors are partitioned 60/11/20 into train, validation and test with sex balance preserved, yielding 4,905/896/1,640 clips. Speaker-independent splitting is not a nicety: a variance decomposition over the feature table gives mean $\eta^2 = 0.178$ for speaker against 0.098 for emotion, with 340 of 436 features speaker-dominant, so a random row split would leak voice identity into the test set and inflate every score below.

Fourteen scikit-learn classifiers [24] spanning tree ensembles [25, 26, 27, 28], kernel machines [29], linear and discriminant models and probabilistic baselines, plus a PyTorch [30] MLP (512–256–128, batch normalisation, dropout 0.3; AdamW, lr $= 10^{-3}$, cosine schedule; class-weighted cross-entropy with label smoothing 0.05; early stopping on validation UAR) — fifteen trained models — are fitted on `ref` and evaluated on all four conditions. Missing values are imputed with training medians and scaling applied only where needed, so tree SHAP values stay in interpretable physical units. The same clips appear in every condition, so a cross-format comparison differs only in format.

For reference throughout: CREMA-D's crowd raters, hearing audio alone, matched the intended emotion on 45.5 % of clips (per-class recall 0.966 for neutral down to 0.164 for sad). Models trained on the *intended* label can exceed this by exploiting acting cues human listeners do not consciously decode; the figure is context for absolute accuracy, not a target.

3.4. Attribution

Attribution is computed on the held-out speakers using TreeSHAP for the ensembles [5], LinearSHAP for the linear models, LIME [3], permutation importance, partial dependence, a depth-4 global surrogate tree, and, for the network, six methods from Captum [8]: Saliency, Input×Gradient, Integrated Gradients, DeepLIFT, GradientSHAP and Feature Ablation. Cross-format stability is Spearman ρ between mean-|attribution| rankings over all 436 features, reported with the number of top-25 features retained.

The RBF-SVM has no exact explainer and was given KernelSHAP with 100 sampled coalitions over 436 features. That least-squares system is rank-deficient and diverged (maximum mean |SHAP| of 2.1×10^{12} against a median of 5.3×10^{-4}). Its KernelSHAP ranking is excluded from every attribution conclusion here; the SVM contributes accuracy and robustness results only.

4. Results

4.1. What the codec does to the features

Fig. 1 characterises the manipulation before any model is involved. Both codecs leave the median feature essentially untouched (median |SMD| 0.020 for MP3, 0.025 for AAC); the difference lives entirely in the tail. Under MP3, 63 of 436 features exceed |SMD| of 0.2 and 28 exceed 0.5, but one descriptor dominates: `spectral_contrast_6_mean`, summarising the 6–8 kHz band, moves by 38.6 training standard deviations against 2.1σ for the next largest. This is perceptual bit allocation seen from the feature side — the encoder abandons the top octave and one column reports it.

AAC's contribution differs in kind. Its largest shifts are all delta-MFCC means (`mfcc_d1_01_mean`, SMD $= -4.16$), consistent with encoder priming shifting frame alignment uniformly:

Table 1: Balanced accuracy on 20 held-out speakers, all models trained on uncompressed audio. Models are grouped by how the decision rule consumes a feature value. Chance $= 0.167$; human voice-only $= 0.455$. `rt` is the container control (AAC re-wrapped as WAV).

	Model	`ref`	`mp3`	Δ	`aac`	`rt`
thresholds	LightGBM	0.591	0.586	−0.005	0.559	0.562
	XGBoost	0.582	0.586	+0.004	0.560	0.560
	HistGB	0.573	0.559	−0.014	0.530	0.533
	Random forest	0.535	0.532	−0.003	0.522	0.522
	AdaBoost	0.521	0.511	−0.010	0.487	0.484
	Extra trees	0.519	0.521	+0.002	0.512	0.507
	Decision tree	0.390	0.382	−0.009	0.367	0.371
multiplies	kNN	0.479	0.362	−0.117	0.467	0.457
	Gaussian NB	0.430	0.277	−0.152	0.437	0.433
	Linear SVM	0.584	0.405	−0.179	0.471	0.462
	MLP	0.574	0.383	−0.191	0.491	0.494
	ANN (PyTorch)	0.599	0.354	−0.246	0.513	0.512
	LDA	0.585	0.338	−0.247	0.507	0.502
	Log. regression	0.567	0.233	−0.334	0.472	0.470
	SVM (RBF)	0.586	0.203	**−0.383**	0.499	0.497

a small perturbation to every first-order derivative rather than the destruction of one band. One caveat belongs with that number — delta means sit near zero with small standard deviations, so their standardised shifts are inflated relative to absolute change.

4.2. In-format performance

Trained and tested on uncompressed audio (the `ref` column of Table 1), the network leads at 0.599 balanced accuracy, with LightGBM, the RBF-SVM, LDA and the linear SVM within two points. Two observations matter, and neither is about the winner. The single decision tree — the one model a human can read end to end — reaches 0.390 against 0.599, so intrinsic interpretability costs a third of the achievable performance here: an argument against Rudin's prescription [10] in this setting and in favour of post-hoc attribution over a strong model. And the spread across the top eight models is under three points, so the leaderboard ordering carries little information. The differences that matter are 12 to 38 points and appear only when the format changes.

4.3. The family split

Table 1 and Fig. 2 give the central result. Under MP3 the outcome is categorical rather than graded. Every tree-based model sits within ±0.014 of its clean score — two of them improve slightly. Every model that consumes features as continuous magnitudes loses between 0.117 and 0.383. There is no overlap between the two groups and no intermediate case; the gap between the worst thresholding model and the best magnitude model is a factor of eight.

The partition does not track accuracy. The best and the worst MP3 performers (the network at 0.599 clean and the decision tree at 0.390) sit on opposite sides of it, and so do the second-best and third-best clean models (LightGBM, −0.005; the RBF-SVM, −0.383). It does not track model capacity, linearity, or whether the model is parametric. It tracks one thing: whether the decision rule compares a feature to a constant or multiplies it by one.

Given §4.1 the mechanism is immediate. A tree asks whether `spectral_contrast_6_mean` exceeds a threshold near 17; the answer is unchanged whether the feature reads 16.4 or 46.4. A linear or kernel model multiplies that 38.6σ excursion straight into its decision function. This is textbook covariate shift [31], with one unusual property: the shifted variable is not a proxy for anything the label depends on, so the shift is pure contamination.

MP4/AAC behaves as its different mechanism predicts — costing everyone something and nobody very much (0.7 to 11.3 points, regardless of family) and leaving the two groups interleaved rather than separated. Whatever governs robustness, then, is an interaction between the codec's error structure and the decision rule, not a property of either alone.

The container is inert. Across the fifteen models the median difference between `roundtrip_wav` and `mp4_aac64` is 0.003 balanced accuracy, with a maximum of 0.010 (kNN), against MP3 codec effects reaching 0.383 — one to two orders of magnitude larger. For a feature pipeline that decodes both routes itself this null is close to structural, and that is precisely its value: confirming it is what licenses attributing everything else to the codec.

Matched-format training is a partial fix. Trained on `mp4_aac64` instead of `ref`, the RBF-SVM scores 0.572 on AAC and logistic regression 0.570, against 0.586 and 0.567 clean-on-clean: the emotional information survives 64 kbps AAC intact, and what fails is the train/serve mismatch. But training on AAC does not rescue MP3 — the SVM falls to 0.167, exactly chance — so the fix is codec-specific, not a general immunisation.

4.4. What the models attribute to

4.4.1. Convergent evidence

On uncompressed audio, five model families and four attribution algorithms select the same acoustic evidence. The top SHAP features for XGBoost and LightGBM are identical in the first five (`mfcc_d1_00_std`, `mfcc_d2_00_std`, `duration_s`, `prosody_intensity_std`, `mfcc_00_std`), and all six gradient methods on the network return the first two plus `prosody_cpps` and `duration_s`. The two leaders are the standard deviations of the first and second derivative of MFCC-0 — how variable the rate of change of overall loudness is, a direct formalisation of vocal agitation. Per-emotion attributions are equally sensible: cepstral peak prominence, a breathiness correlate, drives sadness and neutrality; F_0 median drives fear; F_2 movement drives disgust. The network was told none of this.

4.4.2. The methods disagree with each other

Fig. 3(a) is the uncomfortable result. SHAP tracks a model's *intrinsic* importance closely for the simpler models ($\rho = 0.984$ for the decision tree, 0.983 for logistic regression), but LIME and permutation importance are close to unrelated to it. On the decision tree, LIME and SHAP rank features at $\rho = 0.008$ — no relationship at all — while still sharing 10 of their top 20. This reproduces the disagreement problem Krishna et al. [20] document, on a model simple enough that no approximation error can be blamed.

A within-family comparison sharpens rather than softens the point: the six gradient methods on the network agree almost perfectly ($\rho \geq 0.936$ pairwise, Integrated Gradients and DeepLIFT at 0.998). High agreement *within* a method family is therefore not evidence of correctness — these methods share a mathematical lineage, and their consensus reflects that lineage. A single-method claim about which acoustic features matter is not reproducible across methods; only features surviving all four independent views should be reported, and here that set is `mfcc_d1_00_std`, `prosody_intensity_std`, `duration_s`, `prosody_cpps` and `prosody_f0_median`.

Two calibrations belong with this result. A depth-4 global surrogate reproduces its target model's predictions only 57–74 % of the time (random forest 73.9 %, LightGBM 63.4 %, XGBoost 62.1 %, SVM 58.7 %, decision tree 58.2 %, logistic regression 57.1 %), so surrogate rules summarise a majority of a model, not a model. And the network's attributions pass the model-

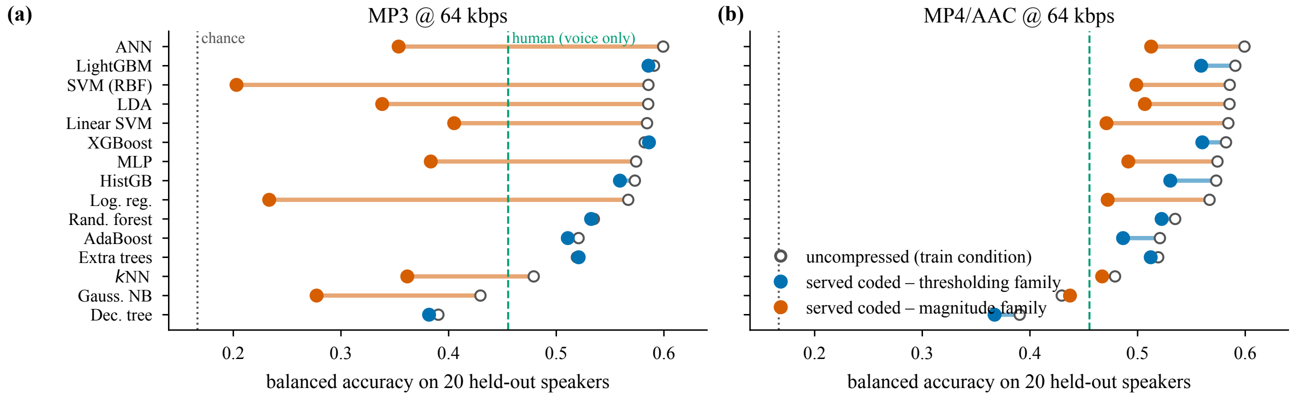

Figure 2: Robustness to lossy coding is determined by how a model consumes its inputs, not by how accurate it is. Open circles are the uncompressed score, filled circles the score when the same fitted model is served coded audio. (a) Under MP3 the two families separate completely: no thresholding model moves more than 0.014, no magnitude model less than 0.117. Four magnitude models fall below the human voice-only ceiling and one lands within 0.04 of chance. (b) Under AAC the same models degrade broadly and mildly, and the families interleave — consistent with a diffuse frame-alignment perturbation rather than a single-band blowout.

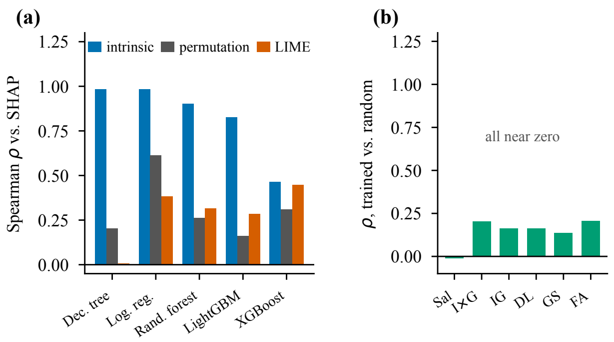

Figure 3: Attribution diagnostics. (a) Post-hoc methods compared against SHAP on the same fitted tabular model, Spearman ρ over all 436 features. SHAP tracks intrinsic importance closely for the simpler models, but LIME and permutation importance are nearly unrelated to it; on the decision tree LIME and SHAP agree at $\rho = 0.008$. (b) Model-randomisation sanity check: all six gradient methods are near zero against a randomly initialised network, so these explanations describe the trained model rather than the input distribution.

randomisation check [19] (Fig. 3(b)): correlation between attributions from the trained network and a randomly initialised one of identical architecture ranges from −0.011 to 0.206 across all six methods.

4.4.3. Stability under coding, and a blind spot

Table 2 gives cross-format attribution stability. The ordering follows the same family split as accuracy: tree and linear attributions barely move under MP3 ($\rho \geq 0.987$), while the network's fall to $\rho = 0.677$ with only 11 of its top 25 features retained.

What replaces what is readable only because the features are named. Under MP3 the features *entering* the network's top 25 are almost all encoder artefacts — `spectral_flatness` in five statistics, plus `spectral_contrast_6_mean` — while the features *leaving* are genuine voice-quality and prosodic measures: `prosody_cpps`, `prosody_f0_skew`, `prosody_f2_mean`, `prosody_f2_std`, `prosody_f3_bw_median`. The compressed model stops listening to the voice and starts listening to the encoder.

The logistic regression row is the most important negative result here. Its SHAP ranking is essentially unchanged under MP3 ($\rho = 0.987$, 22 of 25 top features retained) while its balanced accuracy collapses from 0.567 to 0.233. The coefficients did not change; only the inputs went out of range, and a global mean-|SHAP| ranking is largely blind to that. **A global attribution-stability metric can report "nothing changed" while the classifier has stopped working.** Any deployment monitoring explanation stability as a proxy for model health must pair it with input-range monitoring or per-instance attribution.

Table 2: Attribution stability: Spearman ρ over all 436 features against the uncompressed condition, with top-25 features retained in parentheses. SHAP for all rows except the ANN, which uses Integrated Gradients. The last column repeats the MP3 accuracy change for comparison.

Model	`mp3_64`	`mp4_aac64`	`rt`	Δ acc. MP3
Decision tree	0.995 (24)	0.988 (21)	0.988 (21)	−0.009
XGBoost	0.991 (25)	0.984 (22)	0.984 (22)	+0.004
Random forest	0.990 (25)	0.976 (25)	0.975 (25)	−0.003
LightGBM	0.988 (23)	0.983 (22)	0.982 (22)	−0.005
Log. regression	0.987 (22)	0.942 (21)	0.941 (21)	**−0.334**
ANN (Int. grad.)	**0.677 (11)**	0.891 (16)	0.887 (16)	−0.246

4.5. Diagnosis and repair

Because the features are named, the failure is not merely observable but addressable. Features were ranked by how far each codec moves them in units of *training* standard deviation, then progressively replaced with training medians in both the compressed and the clean test set — the clean column controlling for whatever real signal the masking itself destroys.

Fig. 4 gives the outcome. The single masked feature is `spectral_contrast_6_mean`. Masking it alone lifts the RBF-SVM from 0.203 to 0.577 and the network from 0.354 to 0.580 — recovering 97.9 % and 92.3 % of their respective losses — and three features bring logistic regression to 0.562 against its 0.567 clean-audio score. The tree models are flat throughout, having had nothing to recover.

Two controls make this a diagnosis rather than a coincidence. On uncompressed audio the identical masking costs nothing out to 20 features — the network's clean control sits at 0.606–0.609, marginally *above* its 0.599 unmasked baseline — so these columns were never load-bearing evidence; they were the channel through which the encoder's fingerprint entered the model. And the AAC case behaves as its diffuse mechanism predicts: no single feature dominates, and recovery is gradual, the network peaking at 10 features masked (0.513 → 0.560) and the SVM at 40 (0.499 → 0.542).

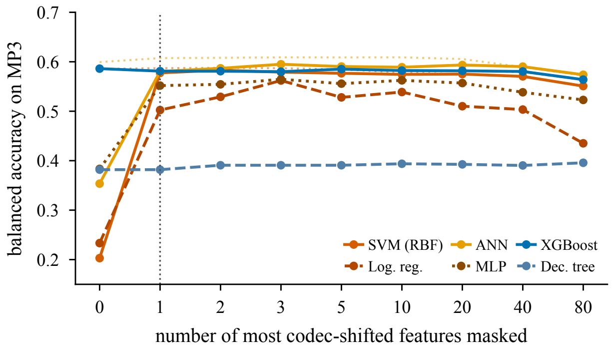

Figure 4: Neutralisation under MP3: replacing the most codec-shifted features with their training medians, ranked by shift in training standard deviations. Masking one feature recovers almost the entire collapse for every affected model; the tree models were never affected and are flat throughout. Beyond roughly 20 features the masking begins to destroy real evidence. Dotted lines show the same masking applied to uncompressed audio, where out to 20 features it costs nothing.

The diagnosis is therefore covariate shift on a small set of identifiable descriptors, not loss of emotional information. The ordering of remedies follows: use a threshold-based model family, which is free; or clamp the top-band spectral descriptors, which recovers almost everything at the cost of one feature; or retrain on the target codec, which works but requires knowing the codec in advance and does not transfer between codecs.

5. Discussion

The practical recommendation is cheap and specific. A SER pipeline that will encounter heterogeneous ingestion formats should either use a thresholding model family, which costs nothing and removes the failure entirely, or monitor and clamp the input range of its most codec-sensitive descriptors. Both are available before any retraining is considered, and both are discoverable only if the features carry acoustic names.

The result also resolves an apparent tension in the literature. Reddy and Vijayarajan [1] report modest, front-end-dependent codec effects on SER; we report catastrophic effects for half our models. Both are correct. The magnitude of a codec effect is not a property of the codec, and reporting it without specifying the model family is reporting an average over a bimodal distribution — which is what our Table 1 shows the distribution to be.

More broadly, explanation stability is a reportable robustness property distinct from accuracy but not a substitute for it. The two move together in rank order here, yet each misses what the other catches: accuracy change consistently *under*-reports attribution change (XGBoost loses 2.2 points on AAC while swapping 3 of its top 25 features; the network loses 8.7 while swapping 9), while in the logistic regression case attribution change dramatically under-reports accuracy collapse. Neither dominates, which is why both belong in a robustness report, alongside the reporting standards urged for ML-based science generally [32].

5.1. Limitations

Results come from a single deterministic actor partition rather than repeated cross-validation, so leaderboard differences of a point or two should not be over-read; the codec effects are 12–38 points and are not at risk from split variance. Only 64 kbps was tested per codec, and the MP3 result is severe partly *because* 64 kbps is aggressive for that codec, so the bitrate at which the family split appears is not established here. Every frame contour is collapsed to order statistics, so no attribution localises *when* in the clip the evidence sits, and attribution comparisons are global — which §4.4 shows can miss a total accuracy collapse. Finally, the corpus is acted English from one dataset, and its 45.5 % human voice-only ceiling shows the labels are only loosely recoverable from audio; absolute accuracies do not transfer to spontaneous speech.

6. Conclusion

Serving a speech emotion model coded audio it was not trained on costs between nothing and two thirds of its headroom above chance, and which of those you get is decided by the model family, not the codec. Under 64 kbps MP3 the split is clean: from bit-identical inputs, thresholding models move by at most 0.014 balanced accuracy and magnitude models by at least 0.117. The cause is a single named descriptor of the 6–8 kHz band arriving 38.6 training standard deviations out of range, and masking that one column recovers almost the entire loss at no cost on clean audio; the container contributes nothing. Two recommendations follow: report the delivery format, an experimental degree of freedom currently going unrecorded, and report codec robustness per model family rather than as an average, because the average describes none of the models it summarises.

7. Data availability and declarations

CREMA-D is public under the terms in [21]. All derived artefacts are released: stimulus scripts and condition manifests, the timestamped 436-feature table over 29,768 rendered files with its schema, the fitted pipelines, and the result files from which every table and figure here is generated; the figure script reads only these and hard-codes no reported value. The study involves no human subjects and no personally identifying data, uses a published corpus of consenting actors released under its authors' own ethical approval, and made no new recordings; SER is dual-use, and these findings are if anything cautionary about deployment without format-aware validation. **CRediT**: A. R — conceptualization, methodology, software, validation, formal analysis, investigation, data curation, writing, visualization. **Conflicts**: none. **Funding**: none; compute costs were borne by the author. **Generative AI** assisted with code, figures and drafting; all design decisions, analyses and interpretations are the author's, every reported value was produced by the committed pipeline, and every citation was verified against a Crossref DOI record or the arXiv API.

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